nRF Connect SDK - latest

NORDIC SEMICONDUCTOR

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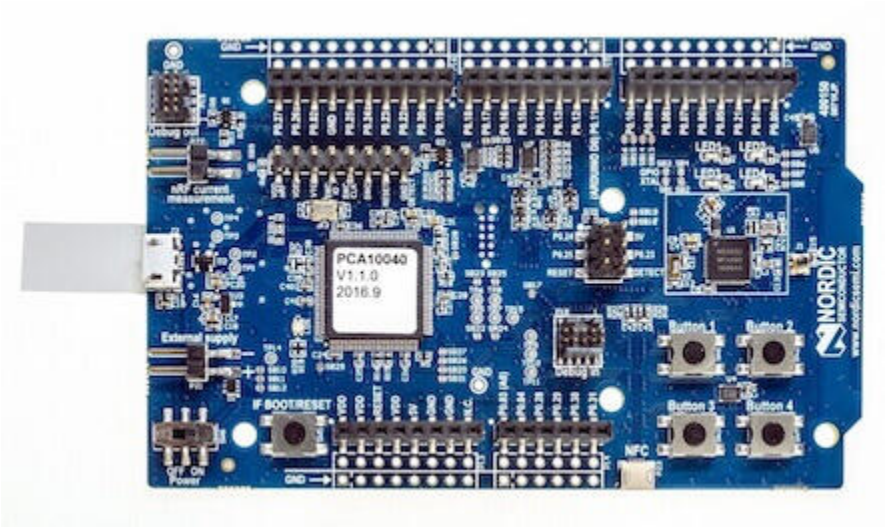
Open on GitHub

Board Overview



nRF52 DK

Name:

`nrf52dk`

Vendor:

Nordic Semiconductor

Architecture:

arm

SoC:

nrf52832, nrf52805, nrf52810

Browse board sources

Overview

The nRF52 Development Kit (PCA10040) hardware provides support for the Nordic Semiconductor nRF52832 ARM Cortex-M4F CPU and the following devices:

- ADC
- CLOCK

- FLASH
- GPIO
- I2C
- MPU
- NVIC
- PWM
- RADIO (Bluetooth Low Energy)
- RTC
- Segger RTT (RTT Console)
- SPI
- UART
- WDT

More information about the board can be found at the nRF52 DK website [1]. nRF52832 Product Specification [2] contains the processor's information and the datasheet.

Hardware

nRF52 DK has two external oscillators. The frequency of the slow clock is 32.768 kHz. The frequency of the main clock is 32 MHz.

Supported Features

The `nrf52dk` board supports the hardware features listed below.

on-chip / **on-board**
Feature integrated in the SoC / present on the board.

2 / **2**
Number of instances that are enabled / disabled. Click on the label to see the first instance of this feature in the board/SoC DTS files.

`vnd,foo`
Compatible string for the Devicetree binding matching the feature. Click on the link to view the binding documentation.

`nrf52dk/nrf52805` target

Type	Location	Description	Compatible
CPU	on-chip	ARM Cortex-M4 CPU1	`arm,cortex-m4`
ADC	on-chip	Nordic Semiconductor nRF family SAADC node1	`nordic,nrf-saadc`
ARM architecture	on-chip	Nordic UICR (User Information Configuration Registers)1	`nordic,nrf-uicr`
	on-chip	Nordic nRF family BPROT (Block Protection)1	`nordic,nrf-bprot`
	on-chip	Nordic EGU (Event Generator Unit)2	`nordic,nrf-egu`
	on-chip	Nordic nRF family SWI (Software Interrupt)4	`nordic,nrf-swi`
Clock control	on-chip	Nordic nRF clock control node1	`nordic,nrf-clock`
	on-chip	Nordic nRF high-frequency crystal oscillator (nRF52 series)1	`nordic,nrf52-hfxo`
Counter	on-chip	Nordic nRF timer node3	`nordic,nrf-timer`
Cryptographic accelerator	on-chip	Nordic ECB (AES electronic codebook mode encryption)1	`nordic,nrf-ecb`
	on-chip	Nordic nRF family CCM (AES CCM mode encryption)1	`nordic,nrf-ccm`
Flash controller	on-chip	Nordic NVMC (Non-Volatile Memory Controller)1	`nordic,nrf52-flash-controller`
GPIO & Headers	on-chip	NRF5 GPIOTE1	`nordic,nrf-gpiote`

Type	Location	Description	Compatible
	on-chip	NRF5 GPIO	`nordic,nrf-gpio`
I2C	on-chip	Nordic nRF family TWI (TWI master)	`nordic,nrf-twi`
Input	on-board	Group of GPIO-bound input keys	`gpio-keys`
Interrupt controller	on-chip	ARMv7-M NVIC (Nested Vectored Interrupt Controller)	`arm,v7m-nvic`
LED	on-board	Group of GPIO-controlled LEDs	`gpio-leds`
Miscellaneous	on-chip	Nordic FICR (Factory Information Configuration Registers)	`nordic,nrf-ficr`
	on-chip	Bluetooth HCI provided by the SoftDevice Controller	`nordic,bt-hci-sdc`
	on-chip	Nordic nRF family PPI (Programmable Peripheral Interconnect)	`nordic,nrf-ppi`
MTD	on-chip	Flash node	`soc-nv-flash`
	on-board	Fixed partitions of a flash (or other non-volatile storage) memory	`fixed-partitions`
Networking	on-chip	Nordic nRF family RADIO peripheral	`nordic,nrf-radio`
Pin control	on-chip	Nordic nRF family Pin Controller	`nordic,nrf-pinctrl`
Power management	on-chip	Nordic nRF power control node	`nordic,nrf-power`

Type	Location	Description	Compatible
PWM	on-chip	nRFx S/W PWM1	`nordic,nrf-sw-pwm`
Regulator	on-chip	Nordic nRF5X regulator (fixed stage of the core supply)1	`nordic,nrf5x-regulator`
Retained memory	on-chip	Nordic GPREGRET (General Purpose Register Retention) device2	`nordic,nrf-gpregret`
RNG	on-chip	Nordic nRF family RNG (Random Number Generator)1	`nordic,nrf-rng`
RTC	on-chip	Nordic nRF RTC (Real-Time Counter)2	`nordic,nrf-rtc`
Sensors	on-chip	Nordic nRF family TEMP node1	`nordic,nrf-temp`
	on-chip	Nordic nRF quadrature decoder (QDEC) node1	`nordic,nrf-qdec`
Serial controller	on-chip	Nordic nRF family UARTE (UART with EasyDMA)1	`nordic,nrf-uarte`
SPI	on-chip	Nordic nRF family SPI (SPI master)1	`nordic,nrf-spi`
SRAM	on-chip	Generic on-chip SRAM1	`mmio-sram`
Timer	on-chip	ARMv7-M System Tick1	`arm,armv7m-systick`
Watchdog	on-chip	Nordic nRF family WDT (Watchdog Timer)1	`nordic,nrf-wdt`

`nrf52dk/nrf52810` target

Type	Location	Description	Compatible
CPU	on-chip	ARM Cortex-M4 CPU1	`arm,cortex-m4`
ADC	on-chip	Nordic Semiconductor nRF family SAADC node1	`nordic,nrf-saadc`
ARM architecture	on-chip	Nordic UICR (User Information Configuration Registers)1	`nordic,nrf-uicr`
	on-chip	Nordic nRF family BPROT (Block Protection)1	`nordic,nrf-bprot`
	on-chip	Nordic EGU (Event Generator Unit)2	`nordic,nrf-egu`
	on-chip	Nordic nRF family SWI (Software Interrupt)4	`nordic,nrf-swi`
Audio	on-chip	Nordic PDM (Pulse Density Modulation interface)1	`nordic,nrf-pdm`
Clock control	on-chip	Nordic nRF clock control node1	`nordic,nrf-clock`
	on-chip	Nordic nRF high-frequency crystal oscillator (nRF52 series)1	`nordic,nrf52-hfxo`
Comparator	on-chip	Nordic nRF COMP (analog COMParator)1	`nordic,nrf-comp`
Counter	on-chip	Nordic nRF timer node3	`nordic,nrf-timer`
Cryptographic accelerator	on-chip	Nordic ECB (AES electronic codebook mode encryption)1	`nordic,nrf-ecb`
	on-chip	Nordic nRF family CCM (AES CCM mode encryption)1	`nordic,nrf-ccm`

Type	Location	Description	Compatible
Debug	on-chip	ARMv7 instrumentation trace macrocell	`arm,armv7m-itm`
Flash controller	on-chip	Nordic NVMC (Non-Volatile Memory Controller)	`nordic,nrf52-flash-controller`
GPIO & Headers	on-chip	NRF5 GPIOTE	`nordic,nrf-gpiote`
	on-chip	NRF5 GPIO	`nordic,nrf-gpio`
I2C	on-chip	Nordic nRF family TWI (TWI master)	`nordic,nrf-twi`
Input	on-board	Group of GPIO-bound input keys	`gpio-keys`
Interrupt controller	on-chip	ARMv7-M NVIC (Nested Vectored Interrupt Controller)	`arm,v7m-nvic`
LED	on-board	Group of GPIO-controlled LEDs	`gpio-leds`
Miscellaneous	on-chip	Nordic FICR (Factory Information Configuration Registers)	`nordic,nrf-ficr`
	on-chip	Bluetooth HCI provided by the SoftDevice Controller	`nordic,bt-hci-sdc`
	on-chip	Nordic nRF family PPI (Programmable Peripheral Interconnect)	`nordic,nrf-ppi`
MTD	on-chip	Flash node	`soc-nv-flash`
	on-board	Fixed partitions of a flash (or other non-volatile storage) memory	`fixed-partitions`

Type	Location	Description	Compatible
Networking	on-chip	Nordic nRF family RADIO peripheral1	`nordic,nrf-radio`
Pin control	on-chip	Nordic nRF family Pin Controller1	`nordic,nrf-pinctrl`
Power management	on-chip	Nordic nRF power control node1	`nordic,nrf-power`
PWM	on-chip	nRF PWM1	`nordic,nrf-pwm`
	on-chip	nRFx S/W PWM1	`nordic,nrf-sw-pwm`
Regulator	on-chip	Nordic nRF5X regulator (fixed stage of the core supply)1	`nordic,nrf5x-regulator`
Retained memory	on-chip	Nordic GPREGRET (General Purpose Register Retention) device2	`nordic,nrf-gpregret`
RNG	on-chip	Nordic nRF family RNG (Random Number Generator)1	`nordic,nrf-rng`
RTC	on-chip	Nordic nRF RTC (Real-Time Counter)2	`nordic,nrf-rtc`
Sensors	on-chip	Nordic nRF family TEMP node1	`nordic,nrf-temp`
	on-chip	Nordic nRF quadrature decoder (QDEC) node1	`nordic,nrf-qdec`
Serial controller	on-chip	Nordic nRF family UARTE (UART with EasyDMA)1	`nordic,nrf-uarte`
SPI	on-chip	Nordic nRF family SPI (SPI master)1	`nordic,nrf-spi`

Type	Location	Description	Compatible
SRAM	on-chip	Generic on-chip SRAM1	`mmio-sram`
Timer	on-chip	ARMv7-M System Tick1	`arm,armv7m-systick`
Watchdog	on-chip	Nordic nRF family WDT (Watchdog Timer)1	`nordic,nrf-wdt`

`nrf52dk/nrf52832` target

Type	Location	Description	Compatible
CPU	on-chip	ARM Cortex-M4F CPU1	`arm,cortex-m4f`
ADC	on-chip	Nordic Semiconductor nRF family SAADC node1	`nordic,nrf-saadc`
	on-board	ADC channels exposed on Arduino Uno (R3) headers1	`arduino,uno-adc`
ARM architecture	on-chip	Nordic UICR (User Information Configuration Registers)1	`nordic,nrf-uicr`
	on-chip	Nordic nRF family BPROT (Block Protection)1	`nordic,nrf-bprot`
	on-chip	Nordic EGU (Event Generator Unit)6	`nordic,nrf-egu`
	on-chip	Nordic nRF family MWU (Memory Watch Unit)1	`nordic,nrf-mwu`
Audio	on-chip	Nordic PDM (Pulse Density Modulation interface)1	`nordic,nrf-pdm`
Clock control	on-chip	Nordic nRF clock control node1	`nordic,nrf-clock`

Type	Location	Description	Compatible
	on-chip	Nordic nRF high-frequency crystal oscillator (nRF52 series)1	`nordic,nrf52-hfxo`
Comparator	on-chip	Nordic nRF COMP (analog COMParator)1	`nordic,nrf-comp`
Counter	on-chip	Nordic nRF timer node5	`nordic,nrf-timer`
Cryptographic accelerator	on-chip	Nordic ECB (AES electronic codebook mode encryption)1	`nordic,nrf-ecb`
	on-chip	Nordic nRF family CCM (AES CCM mode encryption)1	`nordic,nrf-ccm`
Debug	on-chip	ARMv7 instrumentation trace macrocell1	`arm,armv7m-itm`
Flash controller	on-chip	Nordic NVMC (Non-Volatile Memory Controller)1	`nordic,nrf52-flash-controller`
GPIO & Headers	on-chip	NRF5 GPIOTE1	`nordic,nrf-gpiote`
	on-chip	NRF5 GPIO1	`nordic,nrf-gpio`
	on-board	GPIO pins exposed on Arduino Uno (R3) headers1	`arduino-header-r3`
I2C	on-chip	Nordic nRF family TWI (TWI master)11	`nordic,nrf-twi`
I2S	on-chip	Nordic I2S (Inter-IC sound interface)1	`nordic,nrf-i2s`
Input	on-board	Group of GPIO-bound input keys1	`gpio-keys`
Interrupt	on-chip	ARMv7-M NVIC (Nested Vectored Interrupt Controller)1	`arm,v7m-nvic`

Type	Location	Description	Compatible
controller			
LED	on-board	Group of GPIO-controlled LEDs[1]	`gpio-leds`
	on-board	Group of PWM-controlled LEDs[1]	`pwm-leds`
Miscellaneous	on-chip	Nordic FICR (Factory Information Configuration Registers)[1]	`nordic,nrf-ficr`
	on-chip	Bluetooth HCI provided by the SoftDevice Controller[1]	`nordic,bt-hci-sdc`
	on-chip	Nordic nRF family PPI (Programmable Peripheral Interconnect)[1]	`nordic,nrf-ppi`
MTD	on-chip	Flash node[1]	`soc-nv-flash`
	on-board	Fixed partitions of a flash (or other non-volatile storage) memory[1]	`fixed-partitions`
Networking	on-chip	Nordic nRF family RADIO peripheral[1]	`nordic,nrf-radio`
	on-chip	Nordic nRF family NFCT (Near Field Communication Tag)[1]	`nordic,nrf-nfct`
Pin control	on-chip	Nordic nRF family Pin Controller[1]	`nordic,nrf-pinctrl`
Power management	on-chip	Nordic nRF power control node[1]	`nordic,nrf-power`
PWM	on-chip	nRF PWM[1][2]	`nordic,nrf-pwm`

Type	Location	Description	Compatible
	on-chip	nRFx S/W PWM1	`nordic,nrf-sw-pwm`
Regulator	on-chip	Nordic nRF5X regulator (fixed stage of the core supply)1	`nordic,nrf5x-regulator`
Retained memory	on-chip	Nordic GPREGRET (General Purpose Register Retention) device2	`nordic,nrf-gpregret`
RNG	on-chip	Nordic nRF family RNG (Random Number Generator)1	`nordic,nrf-rng`
RTC	on-chip	Nordic nRF RTC (Real-Time Counter)3	`nordic,nrf-rtc`
Sensors	on-chip	Nordic nRF family TEMP node1	`nordic,nrf-temp`
	on-chip	Nordic nRF quadrature decoder (QDEC) node1	`nordic,nrf-qdec`
Serial controller	on-chip	Nordic nRF family UARTE (UART with EasyDMA)1	`nordic,nrf-uarte`
SPI	on-chip	Nordic nRF family SPI (SPI master)21	`nordic,nrf-spi`
SRAM	on-chip	Generic on-chip SRAM1	`mmio-sram`
Timer	on-chip	ARMv7-M System Tick1	`arm,armv7m-systick`
Watchdog	on-chip	Nordic nRF family WDT (Watchdog Timer)1	`nordic,nrf-wdt`

See nRF52 DK website [1] and nRF52832 Product Specification [2] for a complete list of nRF52 Development Kit board hardware features.

Connections and IOs

LED

- LED1 (green) = P0.17
- LED2 (green) = P0.18
- LED3 (green) = P0.19
- LED4 (green) = P0.20
- LD5 (red/green) = OB LED 1/2

Push buttons

- BUTTON1 = SW1 = P0.13
- BUTTON2 = SW2 = P0.14
- BUTTON3 = SW3 = P0.15
- BUTTON4 = SW4 = P0.16
- BOOT = SW5 = boot/reset

External Connectors

J-Link Prog Connector

PIN #	Signal Name
1	VDD
2	IMCU_TMSS
3	GND
4	IMCU_TCKS
5	V5V

PIN #	Signal Name
6	IMCU_TDOS
7	Cut off
8	IMCU_TDIS
9	Cut off
10	IMCU_RESET

Debug IN

PIN #	Signal Name	NRF52832 Functions
1	VDD	N/A
2	SWDIO	SWDIO
3	GND	N/A
4	SWDCLK	SWDCLK
5	GND	N/A
6	P0.18	P0.18 / TRACEDATA[0] / SWO
7	Cut off	N/A

PIN #	Signal Name	NRF52832 Functions
8	Cut off	N/A
9	GND	N/A
10	P0.21	P0.21 / RESET

Debug OUT

PIN #	Signal Name
1	EXT_VTG
2	EXT_SWDIO
3	GND
4	EXT_SWDCLK
5	GND
6	EXT_SWO
7	Cut off
8	Cut off
9	EXT_GND_DETECT

PIN #	Signal Name
10	EXT_RESET

Shield Debug and Current measurement

PIN #	Signal Name
1	VDD_nRF
2	VDD
3	SH_VTG
4	SH_SWDIO
5	SH_SWDCLK
6	SH_SWO
7	SH_RESET
8	SH_GND_DETECT

Auxiliary

PIN #	Signal Name	NRF52832 Functions
1	P0.00	P0.00 / XL1
2	P0.01	P0.01 / XL2

PIN #	Signal Name	NRF52832 Functions
3	P0.21	P0.21 / RESET
4	P0.05_C	P0.05 / AIN3
5	P0.06_C	P0.06
6	P0.07_C	P0.07
7	P0.08_C	P0.08
8	P0.09	P0.09 / NFC1
9	P0.10	P0.10 / NFC2

Arduino Headers

P1/P7 Power

PIN #	Signal Name	NRF52832 Functions
1	VDD	N/A
2	VDD	N/A
3	RESET	P0.21 / RESET
4	VDD	N/A
5	V5V	N/A

PIN #	Signal Name	NRF52832 Functions
6	GND	N/A
7	GND	N/A
8	VIN	N/A

P2/P8 Analog in

PIN #	Signal Name	NRF52832 Functions
1	A0	P0.03 / AIN1
2	A1	P0.04 / AIN2
3	A2	P0.28 / AIN4
4	A3	P0.29 / AIN5
5	A4	P0.30 / AIN6
6	A5	P0.31 / AIN7

P3/P9 Digital I/O

PIN #	Signal Name	NRF52832 Functions
1	D0 (RX)	P0.11
2	D1 (TX)	P0.12

PIN #	Signal Name	NRF52832 Functions
3	D2	P0.13
4	D3	P0.14 / TRACEDATA[3]
5	D4	P0.15 / TRACEDATA[2]
6	D5	P0.16 / TRACEDATA[1]
7	D6	P0.17
8	D7	P0.18 / TRACEDATA[3] / SWO

P4/P10 Digital I/O

PIN #	Signal Name	NRF52832 Functions
1	D8	P0.19
2	D9	P0.20 / TRACECLK
3	D10 (SS)	P0.22
4	D11 (MOSI)	P0.23
5	D12 (MISO)	P0.24
6	D13 (SCK)	P0.25

PIN #	Signal Name	NRF52832 Functions
7	GND	N/A
8	AREF	P0.02 / AIN0
9	SDA	P0.26
10	SCL	P0.27

P5/P11

PIN #	Signal Name	NRF52832 Functions
1	D12 (MISO)	P0.24
2	V5V	N/A
3	D13 (SCK)	P0.25
4	D11 (MOSI)	P0.23
5	RESET	N/A
6	N/A	N/A

Programming and Debugging

The `nrf52dk` board supports the runners and associated west commands listed below.

	flash	debug	debugserver	rtt	attach
jlink	◇	◇ (default)	◇	◇	◇

	flash	debug	debugserver	rtt	attach
nrfjprog	◇				
nrfutil	◇ (default)				
openocd	◇	◇	◇	◇	◇
pyocd	◇	◇	◇	◇	◇

Flashing

Follow the instructions in the Nordic nRF5x Segger J-Link page to install and configure all the necessary software. Further information can be found in Flashing. Then build and flash applications as usual (see Building an Application and Run an Application for more details).

Here is an example for the Hello World application.

First, run your favorite terminal program to listen for output.

```
$ minicom -D <tty_device> -b 115200
```

Replace `<tty_device>` with the port where the board nRF52 DK can be found. For example, under Linux, `/dev/ttyACM0`.

Then build and flash the application in the usual way.

```
# From the root of the zephyr repository
west build -b nrf52dk/nrf52832 samples/hello_world
west flash
```

Debugging

Refer to the Nordic nRF5x Segger J-Link page to learn about debugging Nordic boards with a Segger IC.

Testing the LEDs and buttons in the nRF52 DK

There are 2 samples that allow you to test that the buttons (switches) and LEDs on the board are working properly with Zephyr:

```
samples/basic/blinky
samples/basic/button
```

You can build and flash the examples to make sure Zephyr is running correctly on your board. The button and LED definitions can be found in boards/nordic/nrf52dk/nrf52dk_nrf52832.dts.

nRF52805 emulation on nRF52 DK

The `nrf52dk/nrf52805` board is a modified version of the nRF52 DK that enforces the limitations imposed by the nRF52805 IC, which is a cost-reduced variant of the original nRF52832. Since Nordic does not offer a development kit for the nRF52805, you can use this board to develop for this IC while using the nRF52 Development Kit (PCA10040).

See nRF52805 website [3] for the official reference on the IC itself.

nRF52810 emulation on nRF52 DK

The `nrf52dk/nrf52810` board variant is a modified version of the nRF52 DK that enforces the limitations imposed by the nRF52810 IC, which is a cost-reduced variant of the original nRF52832. Since Nordic does not offer a development kit for the nRF52810 you can use this board to develop for this IC while using the nRF52 Development Kit (PCA10040).

See nRF52810 website [4] for the official reference on the IC itself.

References

[1] (1,2)

https://www.nordicsemi.com/Software-and-Tools/Development-Kits/nRF52-DK

[2] (1,2)

https://docs.nordicsemi.com/bundle/ps_nrf52832/page/nrf52832_ps.html

[3]

https://www.nordicsemi.com/Products/Low-power-short-range-wireless/nRF52805

[4]

https://www.nordicsemi.com/Products/Low-power-short-range-wireless/nRF52810

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